Global Health: Increasing Accessibility and Affordability of Preventative Healthcare

"The world's biggest killer is ischaemic heart disease, responsible for 13% of the world's total deaths."[1] The alarmingly high number of deaths caused by chronic diseases shows the low use of preventative healthcare. Consequently, human quality of life is significantly impaired due to potential loss of income, functional limitations, and broken social relationships.[2] To address the prevalent absence of preventative healthcare, we should move away from private health insurance-based systems, focus less on reactive care, and incorporate artificial intelligence.

Ischaemic heart disease, along with the estimated 80% of cardiovascular disease, is highly preventable.[3] Clinical preventative healthcare is available for many chronic diseases, such as cardiovascular diseases, through three strategies: "intervening before disease occurs (primary prevention), detecting and treating disease at an early stage (secondary prevention), and managing disease to slow or stop its progression (tertiary prevention)."[4] It has been found that greater use of proven clinical preventive services in the United States could save more than two million life-years annually.[5] Yet, as of 2015, only 8 percent of U.S. adults ages 35 and older had received all these recommended preventive services.[6] If preventative healthcare is so effective, why is it substantially underutilized even in developed countries?

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[1] "Global Health Estimates: Leading Causes of Death," n.d., https://www.who.int/data/gho/data/themes/mortality-and-global-health-estimates/ghe-leading-causes-of-death.

[2] Catherine Jane Golics et al., "The Impact of Patients&Amp;#39; Chronic Disease on Family Quality of Life: An Experience From 26 Specialties," *International Journal of General Medicine*, September 1, 2013, 787, https://doi.org/10.2147/ijgm.s45156.

[3] World Heart Federation, "CVD Prevention | What We Do | World Heart Federation," January 25, 2024, https://world-heart-federation.org/what-we-do/prevention/.

[4] Susan Levine et al., "Health Care Industry Insights: Why the Use of Preventive Services Is Still Low," *Preventing Chronic Disease* 16 (March 8, 2019), https://doi.org/10.5888/pcd16.180625.

[5] Michael V. Maciosek et al., "Greater Use of Preventive Services in U.S. Health Care Could Save Lives at Little or No Cost," *Health Affairs* 29, no. 9 (September 1, 2010): 1656–60, https://doi.org/10.1377/hlthaff.2008.0701.

[6] Amanda Borsky et al., "Few Americans Receive All High-Priority, Appropriate Clinical Preventive Services," *Health Affairs* 37, no. 6 (June 1, 2018): 925–28, https://doi.org/10.1377/hlthaff.2017.1248.

The problem is with affordability. There have been past efforts made to decrease financial barriers to accessing preventative healthcare. The most notable instance is the Affordable Care Act (ACA) enacted in March 2010, which required "health plans to cover recommended preventive services without charging a deductible, copayment or co-insurance."[7] The use of recommended preventive services has increased in areas such as cholesterol checks, blood pressure checks, and flu vaccinations for the privately insured population.[8] However, even after the ACA, individuals in the United States still pay for preventative healthcare from their own pockets, with a study stating that "patients are still paying between $75 million and $219 million annually."[9] It is evident that there are still substantial financial barriers to accessing preventative healthcare, especially for those without private insurance and those in countries without policies similar to the ACA.

One major barrier to accessible preventative healthcare is the reliance on private insurance systems. In the United States, where healthcare is predominantly based on private insurance, only individuals with adequate coverage can benefit from policies like ACA. However, approximately 34.6% of the U.S. population—116 million people—do not have private insurance, limiting their access to preventive services.[10] This results in a lower percentage of patients using preventive services in the U.S. compared to countries with healthcare systems not based on private insurance. For example, one study finds that "more than 50% of

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[7] "Background: The Affordable Care Act's New Rules on Preventive Care | CMS," n.d., https://www.cms.gov/cciio/resources/fact-sheets-and-faqs/preventive-care-background.
[8] Xuesong Han et al., "Has Recommended Preventive Service Use Increased After Elimination of Cost-sharing as Part of the Affordable Care Act in the United States?," *Preventive Medicine* 78 (July 23, 2015): 85–91, https://doi.org/10.1016/j.ypmed.2015.07.012.
[9] Alex Hoagland and Paul Shafer, "Out-of-pocket Costs for Preventive Care Persist Almost a Decade After the Affordable Care Act," *Preventive Medicine* 150 (June 15, 2021): 106690, https://doi.org/10.1016/j.ypmed.2021.106690.
[10] US Census Bureau, "Health Insurance Coverage in the United States: 2023," Census.gov, September 11, 2024, https://www.census.gov/library/publications/2024/demo/p60-284.html.

preschoolers [in countries such as Japan, France, Germany, and the United Kingdom] receive preventive care, compared to 42% or less in the United States."[11] By moving away from systems predominantly based on private insurance coverage, countries can promote more equitable access to preventive services. This would decrease mortality rates and improve quality of life (Figure 1). Public health insurance makes screenings, immunizations, and early interventions more accessible and affordable without the financial burden associated with private insurance.

EXHIBIT 3

Effects Of Local Public Health Spending On Community Mortality Rates

Mortality rate	**Percent change per 10% increase in spending**
Infant deaths per 1,000 live births	−6.85***
Heart disease deaths per 100,000 population	−3.22**
Diabetes deaths per 100,000 population	−1.44**
Cancer deaths per 100,000 population	−1.13**
Influenza deaths per 100,000 population	−0.25
All-cause deaths per 100,000 population	−0.29
Alzheimer's deaths per 100,000 population	0.25
Residual deaths per 100,000 population	0.18

Figure 1. Mortality rates due to heart disease (a chronic disease) fell to 3.22% for each 10% increase in local public health spending.[12]

Another critical step is shifting the focus of healthcare systems from reactive treatment to preventative healthcare. Currently, providers are paid mainly to address existing conditions rather than to prevent chronic diseases.[13] This misalignment in financial incentives contributes to underutilizing preventive services, even though providers understand their benefits. For instance, a federal initiative compensating healthcare providers for offering preventive services such as routine screenings and early interventions could close the implementation gap.

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[11] "Preventive Health Care in Six Countries: Models for Reform?," PubMed, n.d., https://pubmed.ncbi.nlm.nih.gov/10138486/.

[12] Glen P. Mays and Sharla A. Smith, "Evidence Links Increases in Public Health Spending to Declines in Preventable Deaths," *Health Affairs* 30, no. 8 (July 22, 2011): 1585–93, https://doi.org/10.1377/hlthaff.2011.0196.

[13] Levine et al., "Health Care Industry Insights: Why the Use of Preventive Services Is Still Low."

Artificial intelligence (AI) can potentially promote the accessibility of preventative healthcare by making it more efficient and affordable. For instance, AI algorithms can monitor wearable devices to detect irregular heart rates or activity levels, alerting individuals to seek medical attention before symptoms escalate.[14] Integrating technology and AI can be more cost-effective for larger populations, making preventative healthcare more accessible for all individuals by removing the need for in-person visits. Using "wearable sensors, biomarkers, and mobile-communications-based education, along with health data collection and analysis, is a viable and affordable way to reach larger populations for better healthcare and compliance."[15] For instance, algorithms can flag abnormalities to ensure timely intervention from health providers for those living in rural areas where it is difficult and costly to access preventive services such as screenings. Thus, healthcare systems can integrate AI and newer technologies into preventive services to increase accessibility and affordability.

The current use of preventive services is alarmingly low, mainly due to financial barriers, and this prevalent issue is costing us millions of lives annually. Ultimately, the accessibility and affordability of preventative healthcare must be promoted through increased public funding and cost-effective AI technologies. Increased government attention and spending on preventative healthcare can be the key to improving human quality of life.

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[14] M Wu, "Wearable Technology Applications in Healthcare: A Literature Review - ProQuest," n.d., https://www.proquest.com/openview/6c96964dfb83ca06895f330233831a50/1?pq-origsite=gscholar&cbl=2034896.
[15] Atam P. Dhawan et al., "Current and Future Challenges in Point-of-Care Technologies: A Paradigm-Shift in Affordable Global Healthcare With Personalized and Preventive Medicine," *IEEE Journal of Translational Engineering in Health and Medicine* 3 (January 1, 2015): 1–10, https://doi.org/10.1109/jtehm.2015.2400919.

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